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Your Roll No.....

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Sr. No. of Question Paper : 1669

Unique Paper Code : 42343307

Name of the Paper : Data Analysis using Python  
Programming (SEC)

Name of the Course : B.Sc. (P) Physical Science  
with Computer Science  
(LOCF)

Semester : III

Duration : 2 Hours

Maximum Marks : 25

**Instructions for Candidates**

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. **Section A** is compulsory.
3. Attempt **any 3** questions from **Section B**.
4. Parts of a question must be answered together.

P.T.O.



## SECTION A

(Compulsory)

1. Consider the following Arrays : (5)

```
arr1 = np.array([[1, 1, 0], [9, 7, 8], [6, 8, 4]])
```

```
arr2 = np.array([[[5, 2, 1], [2, 1, 8]], [[1, 2, 3], [4, 5, 6]]])
```

Find output of the following :

(a) arr2[0]

(b) arr2[1][1]

(c) arr1[:2, -1]

(d) arr1[:,2:]

(e) arr1[-1,:]

2. Consider the following Data Frame : (1+2+1+1=5)

```
Data = {'State': ['Delhi', 'Mumbai', 'Bangalore',  
                 'Delhi', 'Kolkata', 'Pune']}
```

```
'Year': [2001, 1992, 1995, 2005, 1992, 2001]
'Population': [5.2, 1.8, 1.5, 7.8, 1.2, 2.5]
}
```

```
df = pd.DataFrame(Data)
```

Answer the following :

- (a) Rename the indexes as "One", "Two", "Three", "Four", "Five", "Six"
- (b) Add another column in the Data Frame "Extra" having values True corresponding to state 'Delhi' else False.
- (c) Name all the states with Population more than 2.
- (d) Drop rows 'Three' and 'Six'.

### SECTION B

*(Attempt any 3 questions)*

*(Parts of a question must be answered together.)*

3. Consider the Dataset given below : (1+2+1+1=5)

P.T.O.



| Name      | Nationality | Sex    | Age | Handedness |
|-----------|-------------|--------|-----|------------|
| Neha      | India       | Female | 24  | Left       |
| Rahul     | India       | Male   | 28  | Right      |
| Katherine | USA         | Female | 32  | Right      |
| Zenitsu   | Japan       | Male   | 16  | Left       |
| Linda     | USA         | Female | 28  | Left       |
| Mikasa    | Japan       | Female | 22  | Right      |
| Aryan     | India       | Male   | 42  | Right      |
| Gabriel   | USA         | Female | 32  | Right      |
| Kiran     | India       | Female | 20  | Left       |
| Shishir   | USA         | Male   | 40  | Right      |

Answer the following using Contingency Table:

- Find frequency distribution between Sex and Handedness
- Find relationship between Sex, Nationality, and Handedness. Also, display total.
- Normalize the table in part 'b' over each column.



- (d) Find the average of age in part 'a'.
4. (a) Explain a few ways to handle missing values. (2)
- (b) Consider the following DataFrame : (3)

```
df = pd.DataFrame([('bird', 2, 2),  
('mammal', 4, np.nan),  
('arthropod', 8, 0),  
('bird', 2, np.nan)],  
index=('falcon', 'horse', 'spider', 'ostrich'),  
columns=('species', 'legs', 'wings'))
```

What will be the output of the following statements?

- (i) `df.mode()`
- (ii) `df.mode(drop_NA = False)`
- (iii) `df.mode(axis = 'columns', numeric_only = True)`



5. Consider the following Series : (5)

SR = pd.Series([8, -2, 6, 4, 3, 0, 6, -1], index = [0, 3, 4, 6, 7, 10, 12, 14]) Find the output of the following :

- (a) SR.rank()
- (b) SR.rank(method = 'first')
- (c) SR.rank(ascending = False, method = 'min')
- (d) SR.reindex(range(6), method = 'ffill')
- (e) SR.replace({-2: np.nan, 6: 0})

6. Consider the following (df): (1+2+2=5)

| S. No | Restaurant Name  | Location        | User Rating | User Review |
|-------|------------------|-----------------|-------------|-------------|
| 1     | Sky Bar          | Hauz Khas       | 3.4         | Good        |
| 2     | Diggin           | Chanakyapuri    | 4.2         | Very Good   |
| 3     | Cafe Dori        | Nehru Place     | 4.0         | Very Good   |
| 4     | Art of Dumplings | Connaught Place | 4.6         | Excellent   |
| 5     | Flying Saucer    | Connaught Place | 4.0         | Very Good   |
| 6     | Imperfecto       | Hauz Khas       | 4.2         | Very Good   |
| 7     | Sutra            | Chanakyapuri    | 4.4         | Excellent   |
| 8     | Dunkin Donuts    | Nehru Place     | 3.8         | Good        |
| 9     | Wrangler         | Connaught Place | 4.8         | Excellent   |
| 10    | Cha Bar          | Connaught Place | 4.2         | Very Good   |



(a) Group the Restaurants according to their locations and find the restaurant with the highest rating.

(b) Make a word cloud of column User Review. Also use 'Bad' and 'Very Bad' for Stopwords.

(c) Plot Box Plot of User Rating in Data Frame df.

7. Consider the following list : (5)

$x1 = [32, 21, 65, 89, 33, 23, 22, 40, 21, 32, 56, 45, 44, 38, 52]$

(a) From 'x1' create an ndarray 'x2' with 5 rows and 3 columns.

(b) Find Mean, Variance, Median of data in x2 along the rows.

(c) Find cumulative sum of rows in 'x2'.

(d) Sort the data in 'x2' along the columns.



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(e) Replace values greater than 40 in 'x2' with 0  
using 'where' function.

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